

QuadHand: A Compact Quadrotor Aerial Manipulator with MRC-SDF-Based Whole-Body Motion Planning

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Motivations

- A larger arm workspace can induce stronger disturbances on the aerial base.
- Close-proximity interaction requires precise robot and object geometry.
- Coupled base-arm motion makes independent planning restrictive.
- We need compact hardware and efficient whole-body trajectory generation.

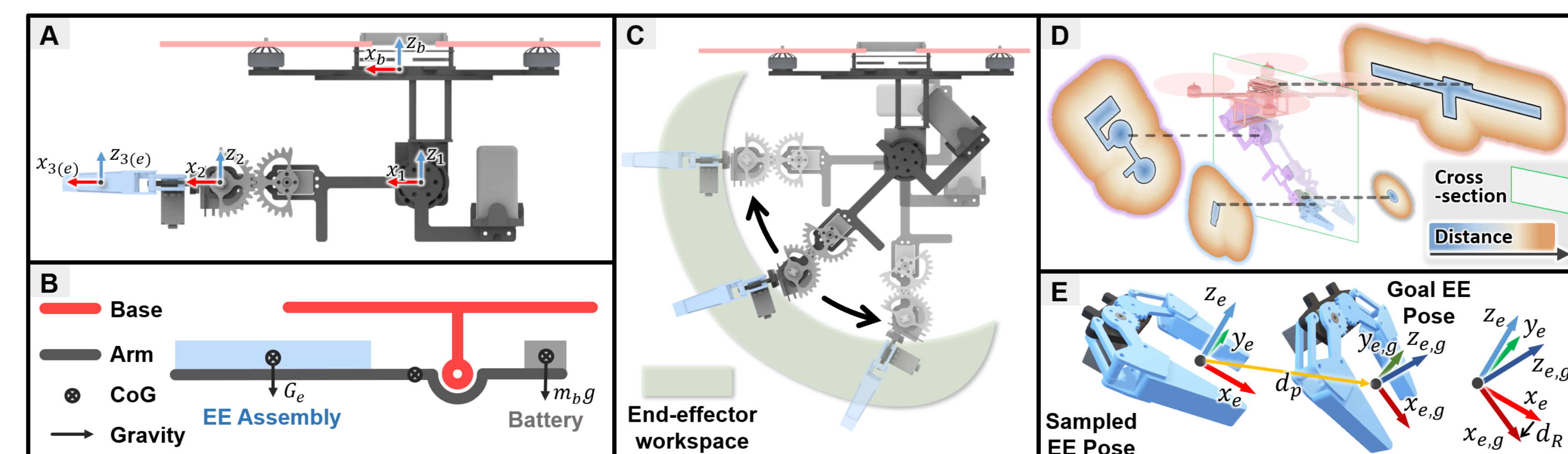


QuadHand: 1.7 kg | 330 mm
3-DoF arm + gripper

Methodology

MRC-SDF: Multi-articulated Robot-Centric Signed Distance Field.

- (A) **Kinematics**: link the base, arm and end-effector coordinate frames.
- (B) **Passive CoG compensation (PCGC)**: battery rotation limits CoG shifts.
- (C) **Workspace**: the 3-DoF arm provides outward end-effector reach.
- (D) **MRC-SDF**: evaluate raw obstacle points with precomputed link-wise SDFs.
- (E) **Insertion**: penalize end-effector position and orientation errors.

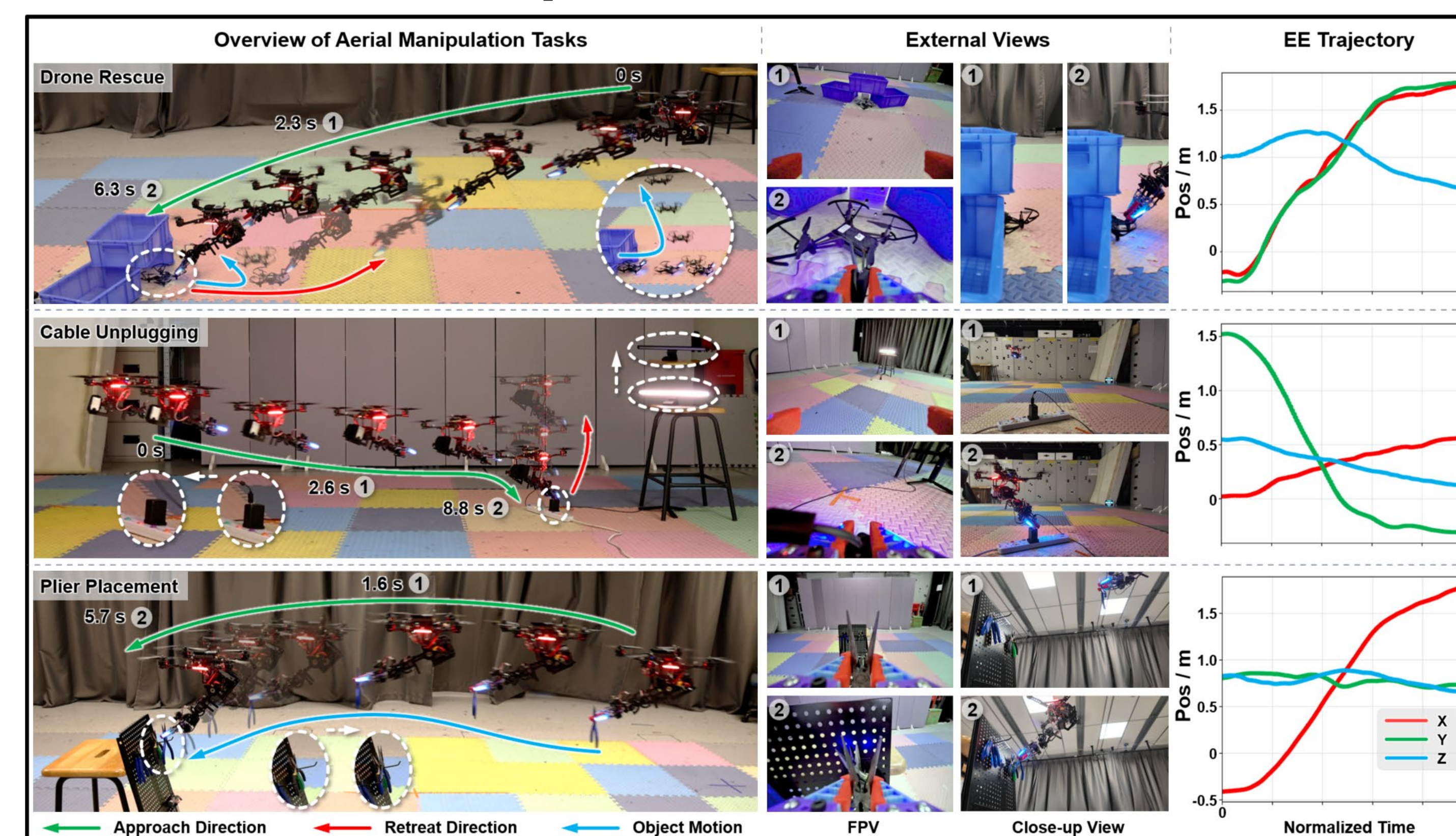


Whole-body MINCO optimization jointly optimizes base motion, arm joints and timing under collision and dynamic-feasibility constraints.

Experiment Result

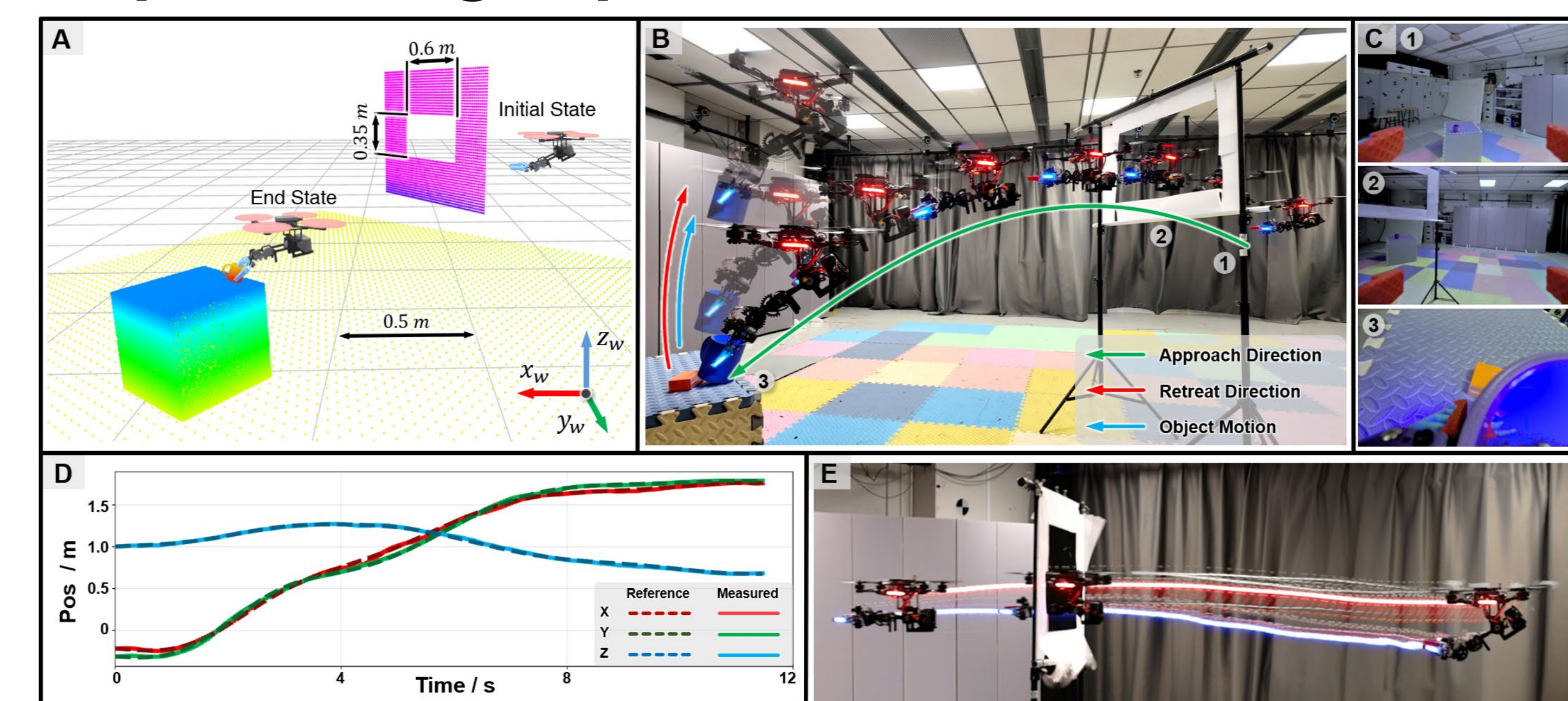
Onboard planning: Jetson Orin NX | State estimation: Vicon + IMU | Terminal states predefined

Real-world manipulation



- ◇ **Drone rescue**: grasp and drag a grounded DJI Tello out from under storage boxes.
- ◇ **Cable unplugging**: grip and remove a USB cable, switching off the powered lamp.
- ◇ **Plier placement**: place a pre-grasped plier onto a hook on a pegboard.

Gap-crossing experiments



- ◇ **Gap-crossing grasp**: cross a 0.60×0.35 m gap and lift a cup with 70/80 mm inner/outer diameter.
- ◇ **Narrow-gap crossing**: 7/7 consecutive trials successfully pass through the gap.
- ◇ **Tracking validation (Fig. 4)**: 3.6×2 m figure-eight, with and without 150 g payload; 3D absolute errors on the order of 3-4 cm. PCGC improves accuracy.